

Sequences and Series

Problem 1

Determine if each of the following sequences converges or diverges. If it converges, find its limit.

(a) $a_n = 2^n$

(f) $a_n = \frac{2^n}{3^n}$

(b) $a_n = (0.2)^n$

(g) $a_n = \frac{2n+1}{n}$

(c) $a_n = (-0.2)^n$

(h) $a_n = \frac{(-1)^n}{n}$

(d) $a_n = 3 + e^{-2n}$

(i) $a_n = \frac{2n+(-1)^n 5}{4n-(-1)^n 3}$

(e) $a_n = 3 + e^{2n}$

(j) $a_n = \frac{\sin(n)}{n}$

Solution.

(a) Since 2^x diverges at infinity so does the sequence $(2^n)_{n \geq 1}$.

(b) Since $|0.2| < 1$, we have $\lim_{n \rightarrow \infty} (0.2)^n = 0$.

(c) Since $|-0.2| < 1$, we have $\lim_{n \rightarrow \infty} (-0.2)^n = 0$.

(d) Since e^{-x} decays at infinity, we have $\lim_{n \rightarrow \infty} (3 + e^{-2n}) = 3$.

(e) Since e^x diverges at infinity so does the sequence with general term $a_n = 3 + e^{2n}$.

(f) Since $|2/3| < 1$, we have $\lim_{n \rightarrow \infty} \frac{2^n}{3^n} = \lim_{n \rightarrow \infty} (2/3)^n = 0$.

(g) Observe that $\lim_{n \rightarrow \infty} \frac{2n+1}{n} = \lim_{n \rightarrow \infty} 2 + \frac{1}{n} = 2$.

(h) Since $\left| \frac{(-1)^n}{n} \right| \leq \frac{1}{n}$, this sequence converges to zero by the squeeze theorem.

(i) Observe that

$$\lim_{n \rightarrow \infty} \frac{2n + (-1)^n 5}{4n - (-1)^n 3} = \lim_{n \rightarrow \infty} \frac{2 + \frac{(-1)^n 5}{n}}{4 - \frac{(-1)^n 3}{n}} = \frac{2}{4} = \frac{1}{2}.$$

(j) Since $\left| \frac{\sin(n)}{n} \right| \leq \frac{1}{n}$, this sequence converges to zero by the squeeze theorem. ■

Problem 2

Consider the sequence $(a_n)_{n \geq 1}$ defined recursively by $a_1 = 1$ and

$$a_n = \sqrt{2 + a_{n-1}}.$$

Write down the first 3 terms of the sequence, prove that it converges and determine its limit.

Solution.

The first 3 terms are given by $a_1 = 1$, $a_2 = \sqrt{3}$, $a_3 = \sqrt{2 + \sqrt{3}}$. Clearly $a_2 \geq a_1$. If we assume that $a_n \geq a_{n-1}$, then

$$a_n \geq a_{n-1} \implies \sqrt{2 + a_n} \geq \sqrt{2 + a_{n-1}} \implies a_{n+1} \geq a_n,$$

so $(a_n)_{n \geq 1}$ is increasing by induction. Furthermore, since $a_1 \leq 2$, if we assume that $a_n \leq 2$, then

$$a_n \leq 2 \implies \sqrt{2 + a_n} \leq \sqrt{2 + 2} \implies a_{n+1} \leq 2,$$

so $(a_n)_{n \geq 1}$ is bounded by 2 by induction. The monotone convergence theorem implies that $(a_n)_{n \geq 1}$ converges to some limit L . Taking the limit on both sides of the recurrence relation reveals that

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \sqrt{2 + a_{n-1}} \implies L = \sqrt{2 + L} \implies L^2 - L - 2 = 0 \implies L = -1, 2.$$

Since $a_n \geq 0$ for all n , we can conclude that $\lim_{n \rightarrow \infty} a_n = 2$. ■

Problem 3

Compute

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)}.$$

Solution.

Fix an integer $m > 1$, and observe that

$$\frac{1}{n(n+1)} = \frac{n+1-n}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}.$$

This means that

$$\sum_{n=1}^m \frac{1}{n(n+1)} = \sum_{n=1}^m \left(\frac{1}{n} - \frac{1}{n+1} \right) = \sum_{n=1}^m \frac{1}{n} - \sum_{n=1}^m \frac{1}{n+1}.$$

Shifting indices and cancelling common terms simplifies this to

$$\sum_{n=1}^m \frac{1}{n(n+1)} = \sum_{n=1}^m \frac{1}{n} - \sum_{n=2}^{m+1} \frac{1}{n} = \frac{1}{1} - \frac{1}{m+1}.$$

Letting m tend to infinity reveals that

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = \lim_{m \rightarrow \infty} \sum_{n=1}^m \frac{1}{n(n+1)} = 1 - \lim_{m \rightarrow \infty} \frac{1}{m} = 1. \quad \blacksquare$$

Problem 4

Find the closed-form for the following geometric series and determine for which values of x they converge.

$$(a) \sum_{n=0}^{\infty} (5x)^n \quad (b) \sum_{n=0}^{\infty} (1-x)^n \quad (c) \sum_{n=2}^{\infty} (1-x)^n \quad (d) \sum_{n=0}^{\infty} \frac{(1-x/2)^n}{2}$$

Solution.

Recall the formula for the geometric series

$$\frac{1}{1-y} = \sum_{n=0}^{\infty} y^n$$

provided that $|y| < 1$.

(a) The formula for the geometric series with $y = 5x$ shows that

$$\sum_{n=0}^{\infty} (5x)^n = \frac{1}{1-5x}$$

for $|5x| < 1$, or equivalently $|x| < 1/5$. This is an interval centered at $x = 0$ of radius $1/5$.

(b) The formula for the geometric series with $y = 1-x$ shows that

$$\sum_{n=0}^{\infty} (1-x)^n = \frac{1}{1-(1-x)} = \frac{1}{x}$$

for $|1-x| < 1$. This is an interval centered at $x = 1$ of radius 1.

(c) Observe that

$$\sum_{n=2}^{\infty} (1-x)^n = \sum_{n=0}^{\infty} (1-x)^n - (1-x) - 1,$$

so part (b) implies that

$$\sum_{n=2}^{\infty} (1-x)^n = \frac{1}{x} + x - 2$$

for $|1-x| < 1$. This is an interval centered at $x = 1$ of radius 1.

(d) Observe that

$$\sum_{n=0}^{\infty} \frac{(1-x/2)^n}{2} = \frac{1}{2} \sum_{n=0}^{\infty} (1-x/2)^n.$$

It follows by the formula for the geometric series that

$$\sum_{n=0}^{\infty} \frac{(1-x/2)^n}{2} = \frac{1}{2} \cdot \frac{1}{1-(1-x/2)} = \frac{1}{x}$$

for $|1-x/2| < 1$, or equivalently $|2-x| < 2$. This is an interval centered at $x = 2$ of radius 2. ■

Problem 5

Bob invests \$200 at the start of each month for 24 months, starting now. If the investment yields 0.5% per month, compounded monthly, what is its value at the end of the 24 months?

Solution.

Bob invests \$200 now, so at the end of the first month he will have

$$200 + 200 \cdot \frac{0.5}{100} = 200 \left(1 + \frac{0.5}{100} \right)$$

dollars. At the start of the next month, he invests another \$200, so at the end of the second month he has

$$\left(200 + 200 \left(1 + \frac{0.5}{100} \right) \right) \cdot \left(1 + \frac{0.5}{100} \right) = 200 \cdot \left(1 + \frac{0.5}{100} \right) + 200 \cdot \left(1 + \frac{0.5}{100} \right)^2.$$

Arguing in this way, it is easy to see that at the end of the n 'th month he has

$$S_n = \sum_{i=1}^n 200 \cdot \left(1 + \frac{0.5}{100} \right)^i$$

dollars. The formula for the geometric partial sums with common ratio $r = 1 + \frac{0.5}{100} = 1.005$ reveals that

$$S_n = \frac{200 \cdot 1.005 \cdot (1 - 1.005^n)}{1 - 1.005}.$$

In particular, at the end of the 24 months period, Bob will have

$$S_{24} = \frac{200 \cdot 1.005 \cdot (1 - 1.005^{24})}{(1 - 1.005)} = 5111.82$$

dollars. ■